LESSON PLAN

PYRO TECHNIQUES (34)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Description on major parts of typical guided missile
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Description on major parts of typical guided missile
5. Specific Objectives	34.1.1.1 Trainee will be able to explain various terms on guided weapon.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of guided missile cutaway diagrams; Safety: Do not handle live components; Instructor supervision: Monitor diagram interpretation accuracy
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define the term 'guided weapon'.
(b) List three major components of a typical guided missile.
(c) Explain the function of the warhead in a missile system.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Propulsion system components • Guidance and control mechanisms • Aerodynamic surfaces function • Warhead integration basics • Missile body structure overview	L & Ds	PPT, IB	15 min
(b)	• Thrust vectoring principles • Inertial navigation systems • Radio command guidance types • Missile casing materials • Ignition sequence procedures	L & Ds	PPT, PDF	20 min
(c)	• Fuzing mechanisms overview • Warhead detonation criteria • Missile separation stages • Control surface actuation • Engine shutdown protocols	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Describe the role of the guidance system in missile flight.
(b) Identify the primary function of aerodynamic surfaces.
(c) Explain the sequence of events from launch to impact.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between active, semi-active, and passive guidance systems for review in the next session on warhead systems.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Warhead system, types and requirements of warhead	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min